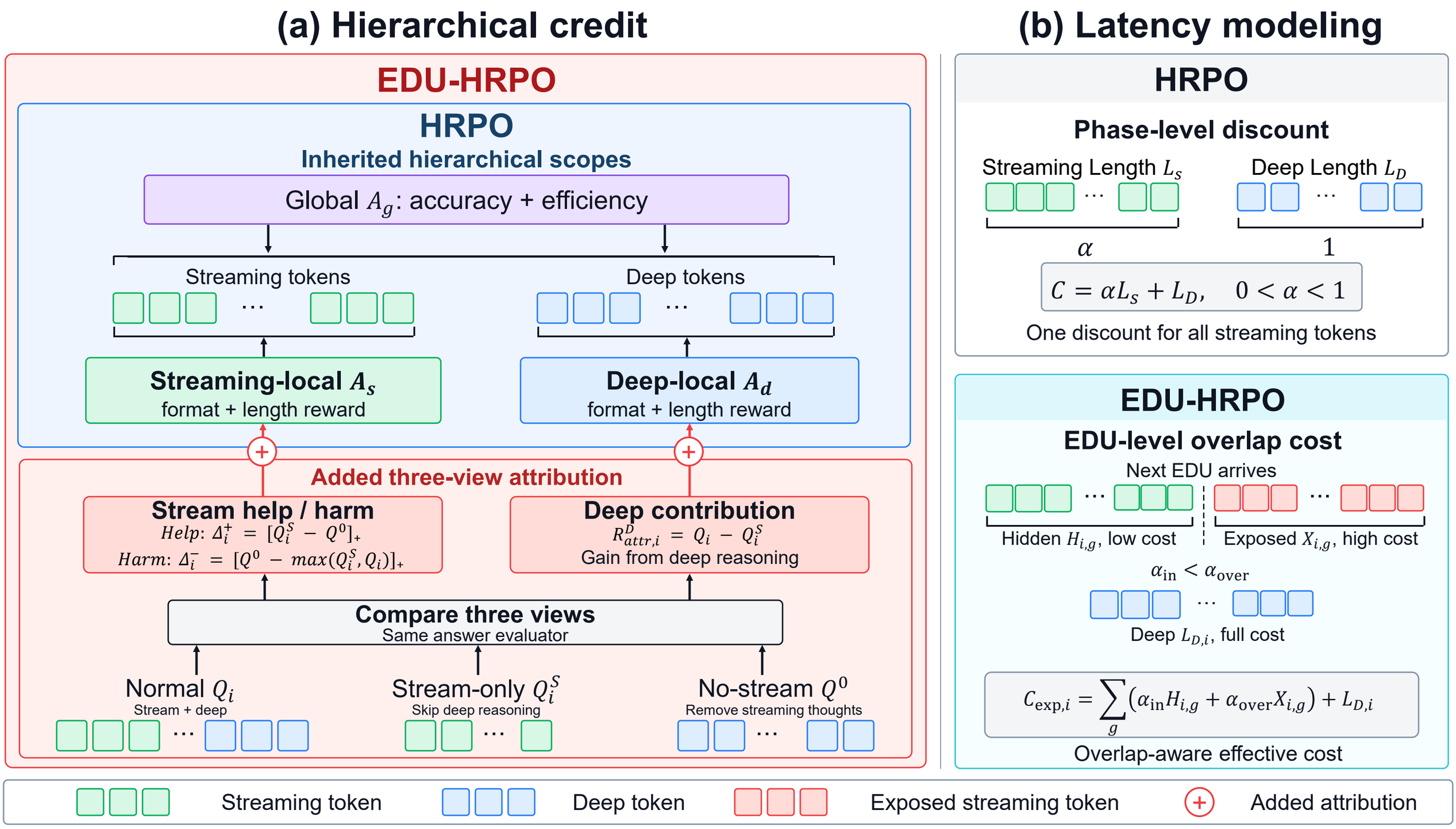

(a) Hierarchical credit
EDU-HRPO
HRPO
Inherited hierarchical scopes
Global $A_g$: accuracy + efficiency
Streaming tokens
Deep tokens
Streaming-local $A_s$
format + length reward
Deep-local $A_d$
format + length reward
Added three-view attribution
Stream help / harm
$Help$: $\Delta_i^+ = [Q_i^S - Q^0]_+$
$Harm$: $\Delta_i^- = [Q^0 - max(Q_i^S, Q_i)]_+$
Deep contribution
$R_{attr,i}^D = Q_i - Q_i^S$
Gain from deep reasoning
Compare three views
Same answer evaluator
Normal $Q_i$
Stream + deep
Stream-only $Q_i^S$
Skip deep reasoning
No-stream $Q^0$
Remove streaming thoughts
(b) Latency modeling
HRPO
Phase-level discount
Streaming Length $L_s$
Deep Length $L_D$
$\alpha$
1
$C = \alpha L_s + L_D, \quad 0 < \alpha < 1$
One discount for all streaming tokens
EDU-HRPO
EDU-level overlap cost
Next EDU arrives
Hidden $H_{i,g}$, low cost
Exposed $X_{i,g}$, high cost
$\alpha_{\text{in}} < \alpha_{\text{over}}$
Deep $L_{D,i}$, full cost
$C_{\text{exp},i} = \sum_g (\alpha_{\text{in}} H_{i,g} + \alpha_{\text{over}} X_{i,g}) + L_{D,i}$
Overlap-aware effective cost
Streaming token
Deep token
Exposed streaming token
Added attribution